GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Junction Box and Cable Gland-BBOU/ELEP/AKAB**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-007
Document Revision	R02
Document Status	Issued for Review
Originator	Paul Roy
Security Classification	Restricted
Issue Date	6-Jul-22

Specification for Junction Box and Cable Gland-BBOU/ELEP/AKAB

R02	07.07.22	Issued for Review	P. Roy	P. Olukah	W.Irem	
R01	26.06.22	Issued for Review	P. Roy	P. Olukah	W.Irem	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	26.06.2022	Issued for Review
R02	07.07.2022	Issued for Review
A01		

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-007

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

Table of Contents

Abbreviations

Abbreviation	Meaning
IEC	International Electro Technical Commission
JB	Junction Box

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant (“GPP”), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

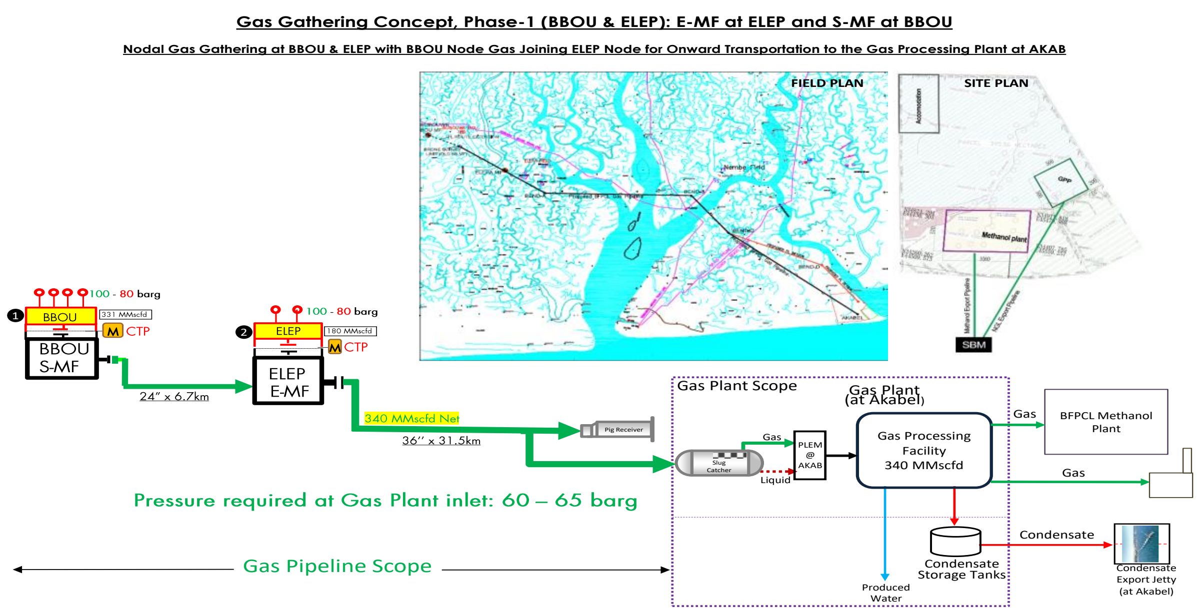

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification defines the minimum requirements for specification for Junction Box and Glands for BBOU, ELEP, AKAB.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Purpose

This specification defines the minimum requirements for specification for Junction Box and Glands for BBOU,ELEP,AKAB.

3.1. General

- All Junction Boxes, Cable Glands shall conform to Shell DEP Standards and shall comply with IEC-600079 standard.
- Each Junction Box shall be provided with a permanently attached tag plate secured to non-removable surface indicating Manufacturer's Name or Trade Mark, Model Number (if any), Serial Number, Purchaser's Junction Box Number and Certification. Tag plate shall be of SS 316 Material.
- The mounting arrangement for Junction Boxes shall be preferably with external mounting lugs suitable for wall, surface, channel or pipe mourning.

3.2. Junction Box Design

- Junction Boxes shall be certified as Ex'e' for intrinsically safe devices and Ex'd' for Explosion Proof devices like Fire and Gas instruments. All instruments shall be certified as IP-65 for Zone1, IIA/IIB, T4.
- The Junction Boxes shall be certified for use in hazardous area as specified. The certification shall be IEC/ATEX/FM/UL.
- The Material of Construction shall be Stainless Steel (SS316).
- Canopy shall be provided for all Junction Boxes.

3.3. Sizing of Junction Box

- Sizing of the Junction Boxes shall be done with due consideration for accessibility and maintenance with the following guidelines:
- Vendor shall suitably size the Junction Box based on the no. of terminals and other design aspects indicated elsewhere in this document.
- Gap of 50 to 60mm between terminals and sides of the Junction Box parallel to the terminal strip for up to 50 terminals and additional 25 mm for each additional 25 terminals.
- Gap of 100 to 120mm between terminals for upto 50 terminals and additional 12 mm for each additional 25 terminals.
- Bottom / Top of terminals shall not be less than 100mm from the bottom /top of the Junction Box
- Alongwith the offer vendor shall supply general arrangement drawing clearly indicating gland arrangements.

3.4. Junction Box Terminals

- Separate Junction boxes shall be provided for AI, DI/DO (24VDC) and F&G signals. For

F&G signals Analog, Digital (24V) and Digital (230 VAC) shall have separate Junction Boxes.

- All Analog and Digital Junctions boxes shall have 24 terminals suitable for 6 Pair X 1.5 mm2 cable main cable and 1P X 1.5 mm2 branch cable. 20% spare terminal shall also be provided.
- All Analog Fire & Gas junction boxes shall have 24 terminals suitable for 6 Triad X 2.5 mm2 main cables and 1 Triad X 1.5 mm2 branch cable. 20% spare terminal shall also be provided.
- The terminals shall be Spring Loaded, Non-Hygroscopic, Vibration Proof Clip-On type mounted on nickel plated steel rail complete with end cover and clamps for each rows. Finish shall mirror type.
- Terminals shall be suitable for minimum 2.5 mm2 multi-stranded copper conductor.
- All terminals shall be provided with suitable markers for identification e.g. Terminal Nos.
- All the terminals shall be preferably DIN rail mounted.
- Each terminal shall be used for connecting single cable, special requirements of series/ parallel connection shall be achieved through special purpose `shorting links/terminals'. Quantity of such links shall be indicated in project specification.
- Terminal colour shall be grey for Explosion proof Junction Box. Intrinsically safe terminals are to be light blue. Terminals shall be mounted on typically Weidmüller (Klippon) type TS-32 rail or similar DIN rail.
- Junction Boxes shall be provided with typically Weidmüller (Klippon) type SAK 2.5 or equivalent makes Phoenix or Wago.
- About 20% spare cable entries and spare terminals shall be provided for each Junction Box. Unused entry holes of JB are to be blanked off with Plugs.

3.5. List of Junction Boxes

Name of Platform	Type of Junction Box	Tag Nos.	Quantity (Nos.)	Branch Cable Size and Gland SIze	Main Cable Size and Gland Size	Remarks
BBOU	Analog	BBOU-AJB-001/002/003/004	4	1 Pair X 1.5 mm2 (½" NPTM)	6 Pair X 1.5 mm2 (¾" NPT)	
BBOU	Digital	BBOU-DJB-001/002/003	3	1 Pair X 1.5 mm2 (½" NPTM)	6 Pair X 1.5 mm2 (¾" NPT)	
BBOU	Fire & Gas	BBOU-FJBA-001, BBOU-FJBD-002	2	1 Triad X 1.5 mm2 (¾" NPT)	6 Triad X 2.5 mm2 (1" NPT)	

Name of Platform	Type of Junction Box	Tag Nos.	Quantity (Nos.)	Branch Cable Size and Gland SIze	Main Cable Size and Gland Size	Remarks
ELEP	Analog	ELEP-AJB-001/002/003/004/005	5	1 Pair X 1.5 mm2 (½" NPTM)	6 Pair X 1.5 mm2 (¾" NPT)	
ELEP	Digital	ELEP-DJB-001/002/003/004	4	1 Pair X 1.5 mm2 (½" NPTM)	6 Pair X 1.5 mm2 (¾" NPT)	
ELEP	Fire & Gas	ELEP-FJBA-001/002, BBOU-FJBD-002	3	1 Triad X 1.5 mm2 (¾" NPT)	6 Triad X 2.5 mm2 (1" NPT)	
AKAB	Analog	AKAB-AJB-001/002	2	1 Pair X 1.5 mm2 (½" NPTM)	6 Pair X 1.5 mm2 (¾" NPT)	
AKAB	Digital	AKAB-DJB-001	1	1 Pair X 1.5 mm2 (½" NPTM)	6 Pair X 1.5 mm2 (¾" NPT)	
AKAB	Fire & Gas	AKAB-FJBA-001 BBOU-FJBD-001	2	1 Triad X 1.5 mm2 (¾" NPT)	6 Triad X 2.5 mm2 (1" NPT)	

Note : Actual Size Junction Box and Gland shall be decided during detail engineering.

3.6. Earthing

- Each Junction Box shall be provided with external earthing lug suitable for 4 mm2 earthing connection. Junction boxes shall have an internal/external stainless steel M10 earthing stud.
- For screen connections of signal cables an electrically isolated copper bar shall be provided along with screws for terminating screens.

3.7. Cable Gland and Plug

- All Cable Glands and plugs shall be supplied along with the Junction Box. The cable entries of junction box shall be on the bottom plate as far as possible, if impossible, then the side plate could be considered, and the top plate is prohibited. Cable glands and plugs shall be SS 316, certified to Exd.
- All Cable Glands shall be double compression type for use with armoured cables.
- The cable glands material shall be Explosion Proof (Exd) of SS 316 material and shall be Weather Proof to IP-65. All unused cable entry points shall be properly plugged with Exd plug & Weather proof to IP-65. ET Gland shall be used for the main cable termination at RTU.

3.8. BOM for Cable Glands and Plug

Note : Actual BOM of Gland shall be decided during detail engineering.

Name of Platform	Gland Size	Certification	Quantity (20% Spare) Nos.
BBOU	½" NPTM	Exd	50
BBOU	¾" NPTM	Exd	25
BBOU	1" NPTM	Exd	4
BBOU	3/4"ET	WP IP-65	25
BBOU	1" ET	WP IP-65	4
ELEP	½" NPTM	Exd	70
ELEP	¾" NPTM	Exd	35
ELEP	1" NPTM	Exd	4
ELEP	3/4"ET	WP IP-65	35
ELEP	1" ET	WP IP-65	4
AKAB	½" NPTM	Exd	24
AKAB	¾" NPTM	Exd	15
AKAB	1" NPTM	Exd	4
AKAB	3/4"ET	WP IP-65	15
AKAB	1" ET	WP IP-65	2